Isaac Fernyhough

isaacfernyhough@gmail.com | 07493 641101 | www.linkedin.com/in/isaac-fernyhough

EDUCATION

MEng Aerospace Engineering, University of Sheffield *Sept 2022 – Jun 2027*

Expected grade: First Class Honours

- **Relevant Modules**: Aircraft Dynamics and Control (95%), State Space Control (85%), Introduction to Systems Engineering and Software (89%), Signals, Systems and Communications (88%), Control Systems Design and Analysis (85%), Machine Learning (77%), Electrical Energy Management (78%), Drone Design, Build and Test (71%)

Durham Johnston Comprehensive *Sept 2020 – Jun 2022*

***A Level** - Maths (A*), Physics (A*), Product Design (A)* ***AS Level** - Further Maths (A), Philosophy and Ethics (A)*

TECHNICAL SKILLS

Programming & Tools: MATLAB, C, Simulink Embedded Coder, Azure DevOps, git, VS Code, Vector Suite (CANalyzer, CANape etc.), Motion Applied toolset (ATLAS, System Monitor, vTAG etc.), FPGAs (VHDL), Microsoft Office

Testing & Verification: Software-in-the-loop (SIL), Hardware-in-the-loop (HIL), Software testing on Dynos

CAD: Fusion 360, Solidworks, Autodesk Inventor

Laboratory Equipment: Oscilloscope, Wave Generator, LabVIEW Software

Manufacturing: Electronics assembly, basic hand-held powered and unpowered tools, carbon fibre layup and post processing, CNC Routing, 3D Printing, laser cutting

Health and Safety: Familiarity with Health and Safety regulations (COSHH etc.) and risk assessments

Driving: Full UK Driving Licence

ENGINEERING EMPLOYMENT EXPERIENCE

Software & Controls Engineering Placement, Mercedes AMG High Performance Powertrains *Sept 2025 – Sept 2026*

Brixworth, Northampton - 12-month Year in Industry

Worked across two motorsports projects in a team of eight software & controls engineers and contributed to dozens of software releases, now deployed and validated in live testing. Gained experience of working collaboratively in a professional and extremely fast-paced environment.

Inverter Software Project

- Led development of Simulink-based inverter control software for the customer's Driver-in-the-loop (DIL) rig; built a MATLAB script that aggregates the required models from the repository into a single model, adds I/O for the customer's VCU signals, and replaces toolbox-dependent blocks with native equivalents - now integrated and in active use for customer testing
- Built a Software-in-the-Loop (SIL) environment with a repeatable MATLAB process to compile the Simulink-based inverter software (excludes FPGA/BIOS code) for Motion Applied's virtual ECU (vTAG) for a build ~70% faster than the main build, enabling faster testing/iteration of the high-level Simulink control software
- Developed a Plant Application with CAN Tx/Rx and a car model to run on the vTAG for the SIL, enabling closed-loop testing of control algorithms and CAN communication ahead of HIL testing
- Developed, integrated and tested the CAN communication software for the Inverter using Simulink and the Vector suite
- Developed, refined and tested torque limiting PID controllers (e.g. DC power and speed limiters), aggregated them into a clear, streamlined torque path and signed off after extensively testing them on the SIL, HIL and Dyno

Vehicle Control Unit (VCU) Software Project

- Developed core vehicle software such as I/O processing, CAN communication, protections and failsafes which continue to help operate the vehicle within its safe running limits
- Resolved faults discovered during track testing by analysing track data using ATLAS, coming up with a solution and delivering it for the next software release
- Worked on the driver interface, including strategy selections using the steering wheel buttons and communication with the dashboard software
- Developed an alternative reverse strategy offering a faster, more flexible way to reverse

ENGINEERING PROJECTS

VTOL Quadplane *Sept 2024 – Sept 2025*
Third-Year Group Project - University of Sheffield

- Avionics Flight Systems lead for a Vertical Take-Off and Landing (VTOL) Quadplane, designed as a reconnaissance tool for hazardous environments; successfully completed an autonomous test flight
- Wired, soldered and integrated all electronics for the quadplane while designing in CAD and 3D printing some components (motor, tail and nose mounts)

Project HEX *Oct 2022 – May 2024*
Extracurricular Group Project - University of Sheffield

- Collaborated with a team of 50 to design, build and fly a 3m wingspan VTOL drone, built to compete in the USA's SUAS competition
- Worked as a manufacturing engineer, programming and operating CNC routing of moulds and created carbon fibre wings and fuselage using wet and dry layup

Quadcopter Drone *Jan 2024 – May 2024*
Second-Year Group Project - University of Sheffield

- Integration lead in a 6-person team for a Quadcopter drone project which showed impressive robustness and lateral stability during the flight demonstration
- Designed the drone in CAD and developed extremely light and robust drone 'legs' using generative design

Autonomous Robot *Sept 2023 – Feb 2024*
Second-Year Group Project - University of Sheffield

- Worked as a team to program a two-wheeled robot to autonomously avoid objects, move to a beacon and stop at the beacon using C, receiving a score of 100% for our robot demonstration

UKSEDS National Rocketry Competition *Oct 2022 – May 2023*
Extracurricular Group Project - University of Sheffield

- Design engineer in a multidisciplinary team, designing, building and launching a rocket
- Competed against other UK universities and finished in 2nd place nationally

OTHER EMPLOYMENT & VOLUNTEERING EXPERIENCE

Young Professional - Events Manager, Institution of Engineering and Technology (IET) *Apr 2022 – Present*

- Organising events (webinars, guest lectures and field trips) that provide insight into real engineering practices and new innovations for like-minded students
- Providing networking opportunities with leading industry professionals for fellow students
- Working collaboratively to find venues and organise food and drinks for events

Research Intern, Innerscape *Jun 2024 – Oct 2024*

- Conducting market research on a variety of video games and evaluating their use of inner experience
- Writing reports and providing briefs for meetings with industry leaders

Part-time work:

- Stall Assistant, Confero Marketing - The Great North Run, Confero Marketing *Sept 2023*
- Bartender, The Victoria Inn *Jun 2021 – Jun 2024*

CHARITY & EXTRA-CURRICULAR ACTIVITIES

Movember Charity

- 2025: Organised Mercedes AMG HPP's efforts where we raised £3,558 as a team, I raised £300 personally and ran 100km during the month
- 2023: Participated as part of University of Sheffield Cricket Club, where I personally raised £365 by running and hiking 150km including completing the Yorkshire Three Peaks

Mercedes AMG HPP Football Team: Competed against other F1 teams

Mercedes AMG HPP Cricket Team: Played in charity games and competed against other F1 teams

University of Sheffield Cricket Club: Performance Team

Bronze Mathematics Award: St John's School and Sixth Form